EXECUTIVE PROPOSAL

Universal Anomaly Engine

Enterprise-Grade Deep Learning Suite for Mission-Critical Detection

Executive Summary
This project introduces the **Universal Anomaly Engine**, a high-fidelity intelligence platform designed to detect structural and behavioral deviations across three core domains: **Asset Management (Finance)**, **Radiology (Healthcare)**, and **Network Defense (Cybersecurity)**. By utilizing unsupervised **PyTorch Autoencoder** architectures, the engine identifies risk with mathematical precision.

🏛️ System Architecture: The "Enterprise Elegant" Framework

To ensure maximum scalability and high-fidelity UX, we utilized a split-architecture:

- **Intelligence Layer (Hugging Face):** Optimized PyTorch model serving with Python 3.10 stabilization.
- **Operational Layer (Vercel):** A sophisticated React dashboard providing real-time telemetry and advanced glassmorphism visuals.

🧠 Core Intelligence Modules

A. Financial Intelligence (Dense AE): Identification of fraudulent transaction vectors using a 30-feature PCA latent space compression.

B. Radiology Diagnostics (Spatial Conv2D AE): Automated detection of Pediatric Pneumonia using Spatial Convolutional layers mapping pixel-level morphology. Pathological scans produce a massive **13x signal** increase in reconstruction error.

C. Cybersecurity Operations (Deep Dense AE): Zero-day intrusion analysis using 77-dimensional entropy scoring, successfully flagging DoS and Neptune attacks with over 94% accuracy.

📈 Platform Maturity & ROI

The Universal Anomaly Engine represents a shift from reactive to **proactive defense**. By learning "Normal" behavior, the engine protects against previously unknown (zero-day) anomalies that traditional rules-based systems miss.

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